ML Lab 10

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We have used UCI data on wine - https://archive.ics.uci.edu/dataset/109/wine

This has a lot of features so we apply LDA for dimensionality reduction

Imports

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import f1_score
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
```

Data Preprocessing and Exploratory Analysis

```
cols = ['Wine_Origin','Alcohol','Malic_acid','Ash','Alcalinity_of_ash','Magnesi
    'Flavanoids','Nonflavanoid_phenols','Proanthocyanins','Color_intensity','Hu

df = pd.read_csv('./data/wine.data',names=cols,index_col=False)

df.head()
```

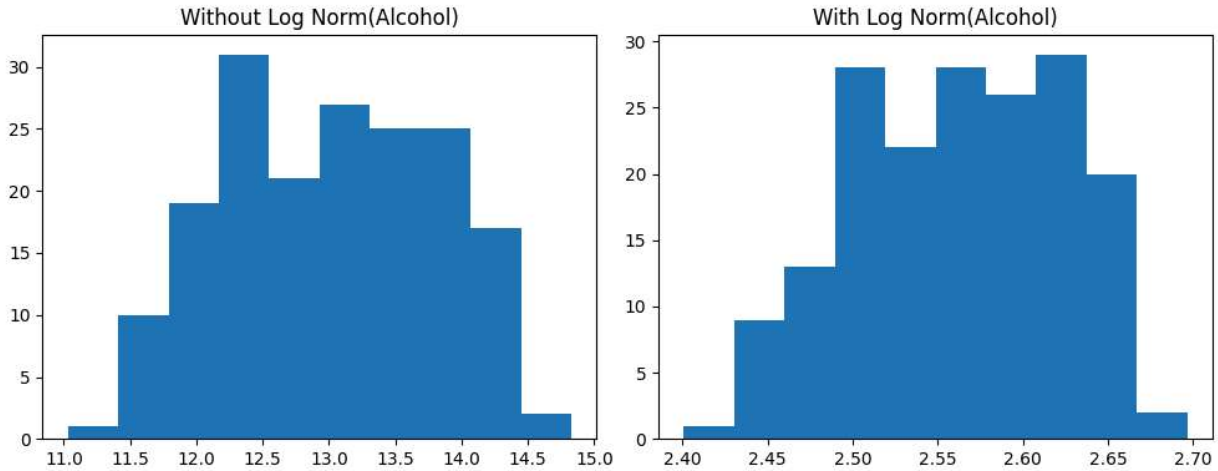
	Wine_Origin	Alcohol	Malic_acid	Ash	Alcalinity_of_ash	Magnesium	Total_ph
0	1	14.23	1.71	2.43	15.6	127	
1	1	13.20	1.78	2.14	11.2	100	
2	1	13.16	2.36	2.67	18.6	101	
3	1	14.37	1.95	2.50	16.8	113	
4	1	13.24	2.59	2.87	21.0	118	

```
df.describe()
```

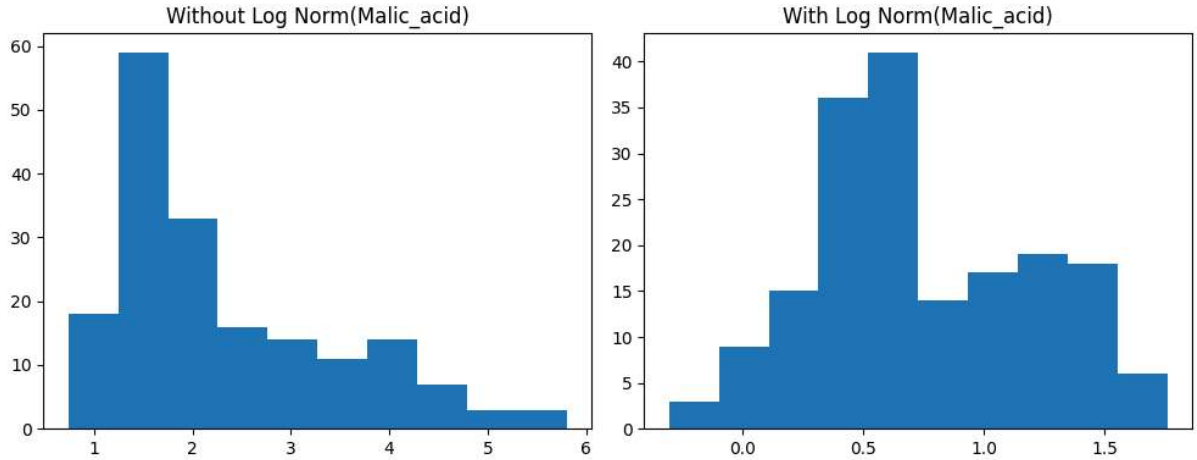
	Wine_Origin	Alcohol	Malic_acid	Ash	Alcalinity_of_ash	Magne
count	178.000000	178.000000	178.000000	178.000000	178.000000	178.00
mean	1.938202	13.000618	2.336348	2.366517	19.494944	99.74
std	0.775035	0.811827	1.117146	0.274344	3.339564	14.28
min	1.000000	11.030000	0.740000	1.360000	10.600000	70.00
25%	1.000000	12.362500	1.602500	2.210000	17.200000	88.00
50%	2.000000	13.050000	1.865000	2.360000	19.500000	98.00
75%	3.000000	13.677500	3.082500	2.557500	21.500000	107.00
max	3.000000	14.830000	5.800000	3.230000	30.000000	162.00

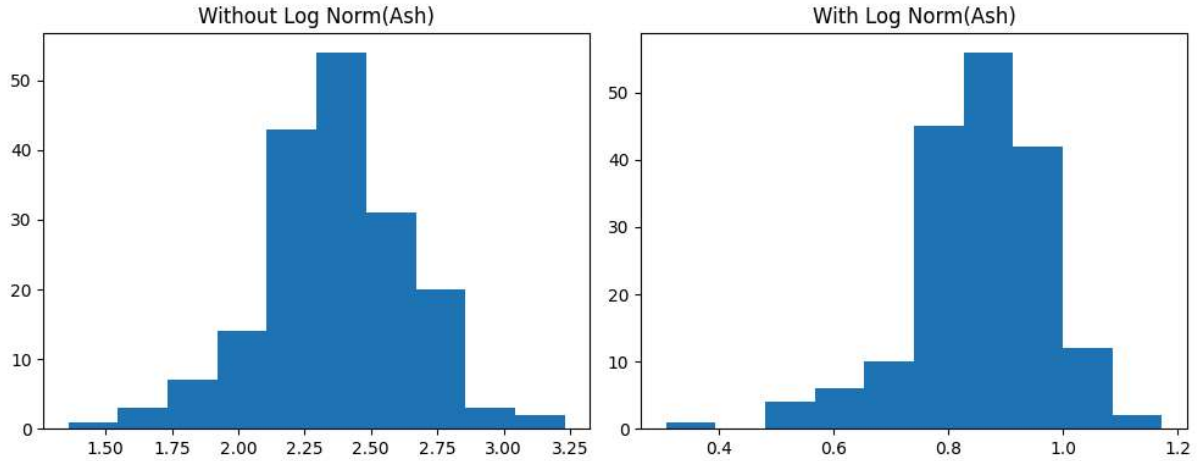
```
def show_diff_and_log(df,col_name):
    plt.figure(figsize=(10, 4))


    plt.subplot(1,2,1)
    plt.title(f'Without Log Norm({col_name})')
    plt.hist(df[col_name])

    plt.subplot(1,2,2)
    plt.title(f'With Log Norm({col_name})')
    plt.hist(np.log(df[col_name]))

    plt.tight_layout()
    plt.show()

    df[col_name] = np.log(df[col_name])
```

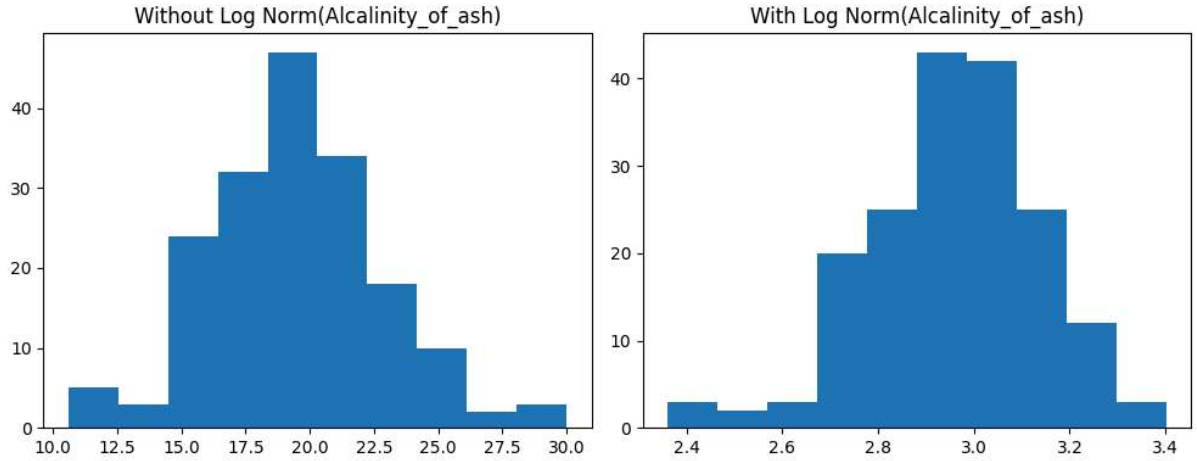
```
show_diff_and_log(df,'Alcohol')
```

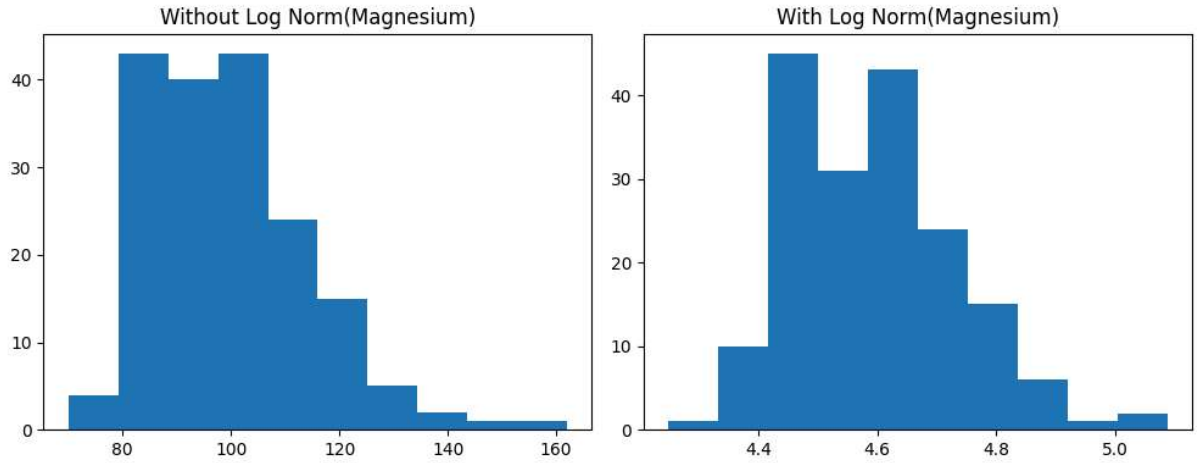



```
to_norm = cols.copy()
to_norm.remove('Alcohol')
to_norm.remove('Wine_Origin')
for col in to_norm:
    show_diff_and_log(df,col)
```

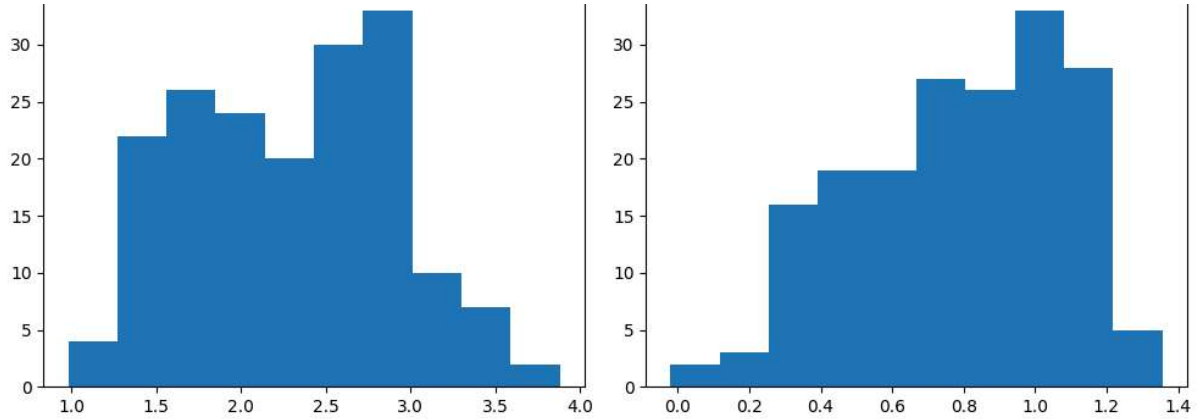

Without Log Norm(Malic_acid)
With Log Norm(Malic_acid)


Without Log Norm(Ash)
With Log Norm(Ash)

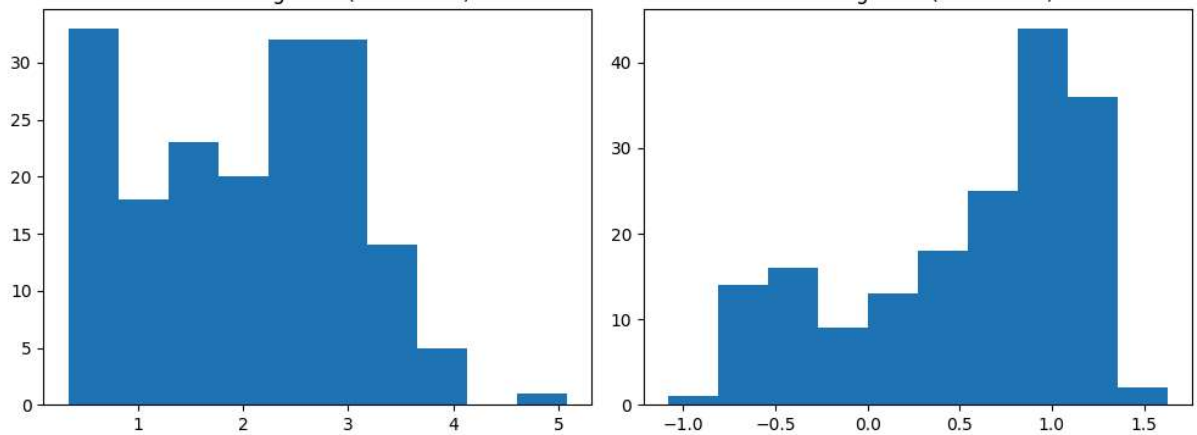

Without Log Norm(Alcalinity_of_ash)
With Log Norm(Alcalinity_of_ash)


Without Log Norm(Magnesium)
With Log Norm(Magnesium)

Without Log Norm(Total_phenols)
With Log Norm(Total_phenols)

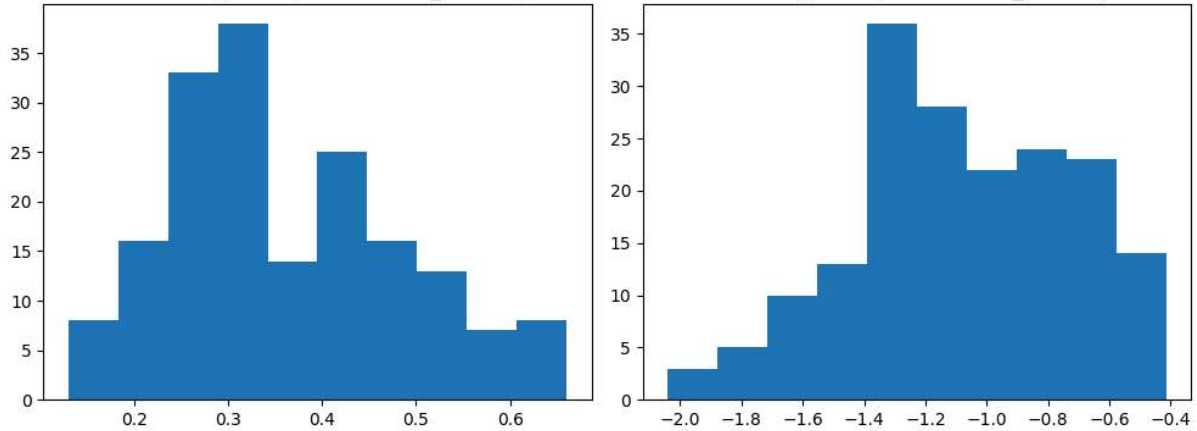


Without Log Norm(Flavanoids)



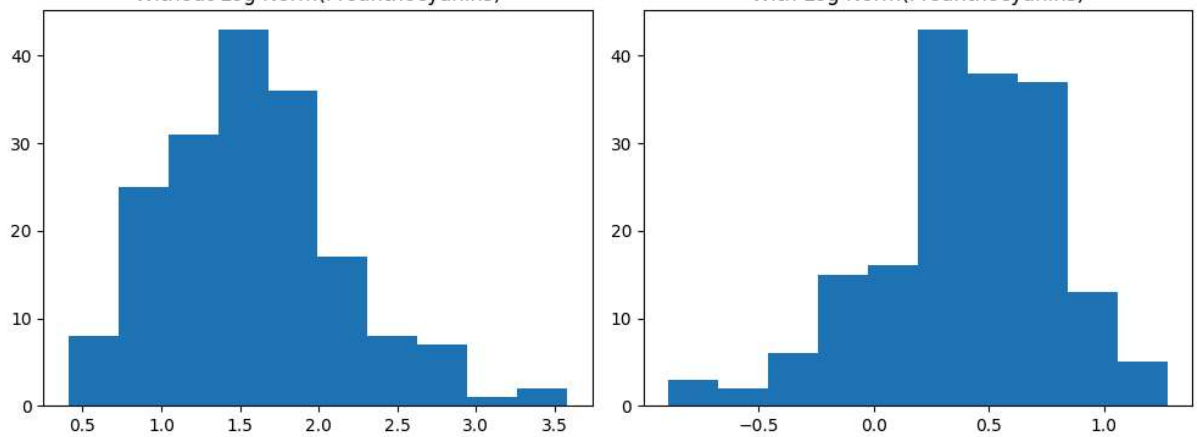
With Log Norm(Flavanoids)

Without Log Norm(Nonflavanoid_phenols)



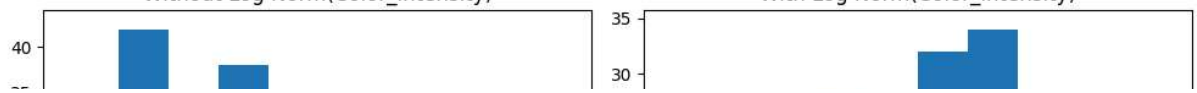
With Log Norm(Nonflavanoid_phenols)

Without Log Norm(Proanthocyanins)



With Log Norm(Proanthocyanins)

Without Log Norm(Color_intensity)



With Log Norm(Color_intensity)